

Initiating Search

November 10, 2025, 9:30 PM

• Search:

CAS Lexicon Search:

Hantzsch cyclocondensation reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (1,465)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,740)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Publication Year:	2020 to 2025	

Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

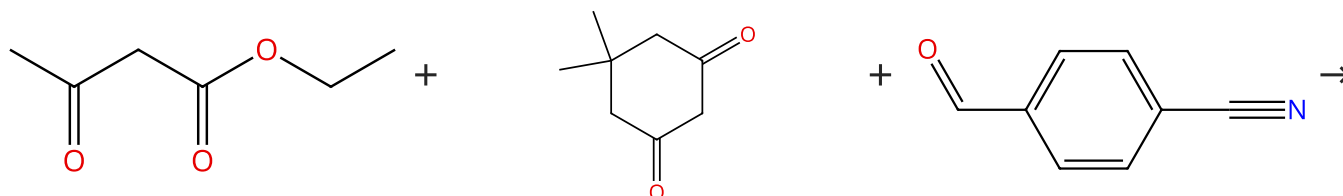
Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.

Reactions (50)

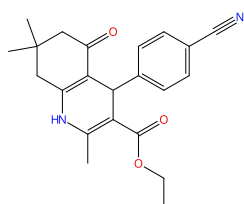
[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (95)

 Suppliers (102)

 Suppliers (91)

 Suppliers (2)

31-614-CAS-39552438

Steps: 1 Yield: 100%

1.1 **Reagents:** Ammonium acetate
Catalysts: Papain (immobilized onto glutaraldehyde-functionalized nitrogen-doped graphene...), Graphene (papain immobilized onto glutaraldehyde-functionalized nitrogen-doped); 2 min

Experimental Protocols

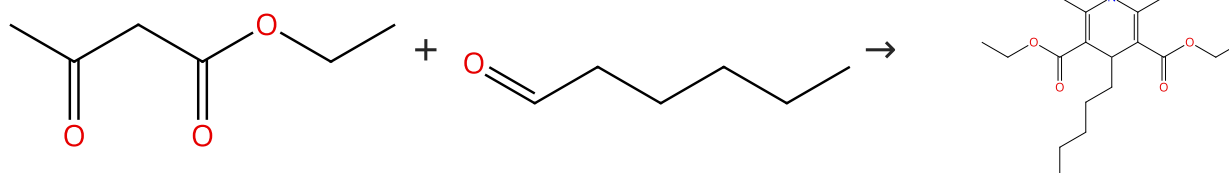
Immobilization of papain onto nitrogen graphene quantum dots and its use as an efficient catalyst for facile MW-assisted synthesis of polyhydroquinolines

By: Mirzaei, Farzaneh; et al

Journal of the Iranian Chemical Society (2024), 21(3), 639-648.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (95)

 Suppliers (68)

 Suppliers (8)

31-614-CAS-31839613

Steps: 1 Yield: 100%

1.1 **Reagents:** Ammonium acetate
Catalysts: Europium trichloride
Solvents: Ethanol; 2 h, 78 °C

Experimental Protocols

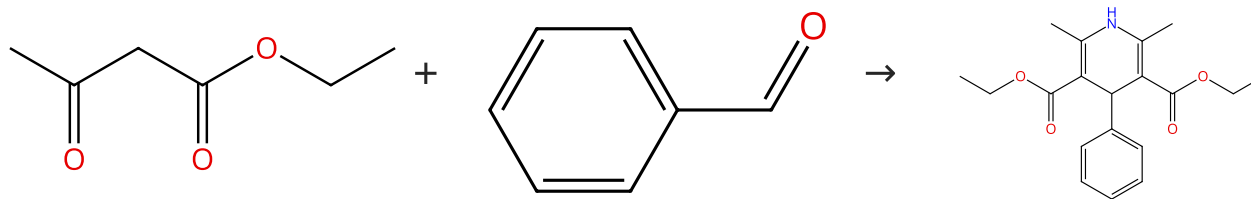
New 1,4-Dihydropyridines. Optimization of the Synthesis and In Silico Analysis of Biological Activity

By: Mineyeva, I. V.; et al

Russian Journal of Organic Chemistry (2022), 58(3), 268-281.

Scheme 3 (7 Reactions)

Steps: 1 Yield: 98-100%



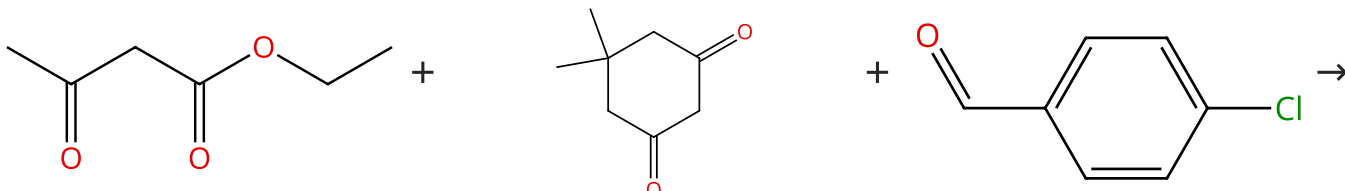
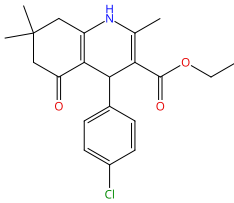
Suppliers (95)

Suppliers (69)

Suppliers (40)

31-614-CAS-43972783 Steps: 1 Yield: 100% 1.1 Reagents: Ammonium acetate Catalysts: 3064002-69-0 Solvents: Ethanol; 4 h, 60 °C Experimental Protocols	Stoichiometry-Regulated Synthesis of Three Adenine-Based Coordination Polymers for Catalytic Excellence through the Synergistic Amalgamation of Coordinative Unsaturation and Lewis Basic Sites By: Patra, Rajesh; et al Inorganic Chemistry (2024), 63(49), 23396-23410.
31-614-CAS-39082056 Steps: 1 Yield: 99% 1.1 Reagents: Ammonium acetate Catalysts: 3027472-69-8 Solvents: Ethanol; 4 h, 60 °C Experimental Protocols	A Highly Chemically Robust 3D Interpenetrated MOF Heterogeneous Catalyst for the Synthesis of Hantzsch 1,4-Dihydropyridines and Drug Molecules By: Sahoo, Rupam; et al Small (2024), 20(25), 2309281.
31-614-CAS-36100079 Steps: 1 Yield: 99% 1.1 Reagents: Ammonium acetate Catalysts: Manganate(4-), chloro[[4,4',4'',4'''-(21<i>H</i>,23<i>H</i>-porphine-5,10,15,20-tetrayl-κ<i>N</i>²¹,κ<i>N</i>²²,κ<i>N</i>²³,κ<i>N</i>²⁴)tetrakis[benzenesulfonato]](6-)]-, hydrogen (1:4), (<i>SP</i>-5-12)- Solvents: Ethanol, Water; 15 min, reflux Experimental Protocols	Sequential Assembly and Stabilization of Cu₆S₆ Octahedral Clusters in NaCl-, NiAs-, and CdI₂-Related Structures and Their Utility toward Thermochromism and Multicomponent Hantzsch Reaction By: Sarkar, Anupam; et al Inorganic Chemistry (2023), 62(11), 4417-4434.
31-333-CAS-23712770 Steps: 1 Yield: 99% 1.1 Reagents: Ammonium acetate Catalysts: Manganate(4-), chloro[[4,4',4'',4'''-(21<i>H</i>,23<i>H</i>-porphine-5,10,15,20-tetrayl-κ<i>N</i>²¹,κ<i>N</i>²²,κ<i>N</i>²³,κ<i>N</i>²⁴)tetrakis[benzenesulfonato]](6-)]-, hydrogen (1:4), (<i>SP</i>-5-12)- Solvents: Ethanol, Water; rt; 15 min, reflux Experimental Protocols	Boosting the catalytic performance of manganese (III)-porphyrin complex MnTSP for facile one-pot green synthesis of 1,4-dihydropyridine derivatives under mild conditions By: Ali El-Remaily, Mahmoud Abd El Aleem Ali; et al Applied Organometallic Chemistry (2021), 35(7), e6238.
31-614-CAS-24995368 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: 2741906-90-9 Solvents: Ethanol; 2 h, 60 °C Experimental Protocols	Intrinsic-Unsaturation-Enriched Biporous and Chemorobust Cu(II) Framework for Efficient Catalytic CO₂ Fixation and Pore-Fitting Actuated Size-Exclusive Hantzsch Condensation with Mechanistic Validation By: Seal, Nilanjan; et al ACS Applied Materials & Interfaces (2021), 13(46), 55123-55135.
31-614-CAS-32141327 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium carbonate; 3 - 4 min Experimental Protocols	Solar energy as a renewable energy source for preparative-scale as well as solvent and catalyst-free Hantzsch reaction By: Gadkari, Yatin U.; et al Sustainable Chemistry and Pharmacy (2021), 21, 100444.

31-333-CAS-23765971	Steps: 1 Yield: 98%	Synthesis, Characterization and Application of β -Cyclodextrin/Imidazolium Based Dicationic Ionic Liquid Supported on Silica Gel as a Novel Catalyst in Hantzsch Condensation Reaction
1.1 Reagents: Ammonium acetate Catalysts: β -Cyclodextrin (reaction products with imidazole and chlorosilane), 1,1'-(1,4-Butanediyl)bis(imidazole) (reaction products with cyclodextrin and chlorosilane), [3-Chloro-2-(chloromethyl)propyl]trimethoxysilane (reaction products with cyclodextrin and imidazole); 15 min, 90 °C		By: Moheiseni, Fatemeh; et al Polycyclic Aromatic Compounds (2021), 41(5), 1094-1106.
Experimental Protocols		

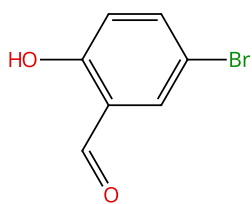
Scheme 4 (11 Reactions)	Steps: 1 Yield: 98-100%
	
	
Suppliers (95)	Suppliers (102)
Suppliers (98)	Suppliers (13)

31-614-CAS-24483434	Steps: 1 Yield: 100%	Introduction of a New Nano Sized Ni-Based Salt for the Acceleration of the Synthesis of Pyrano [2,3-d]pyrimidinone and 1,4-Dihydropyridine Derivatives
1.1 Reagents: Ammonium acetate Catalysts: 2768645-88-9; 7 min, 120 °C		By: Yarinia, Ramtin; et al Polycyclic Aromatic Compounds (2022), 42(10), 6795-6809.
Experimental Protocols		
31-333-CAS-21680965	Steps: 1 Yield: 100%	Introduction of an efficient DABCO-based bis-dicationic ionic salt catalyst for the synthesis of arylidenemalononitrile, pyran and polyhydroquinoline derivatives
1.1 Reagents: Ammonium acetate Catalysts: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1,1'-(2-hydroxy-1,3-propanediyl)bis-, chloride (1:2); 10 min, 120 °C		By: Zabihzadeh, Mehdi; et al Journal of Molecular Structure (2020), 1206, 127730.
Experimental Protocols		
31-333-CAS-21579510	Steps: 1 Yield: 100%	New Copper Complex on Fe ₃ O ₄ Nanoparticles as a Highly Efficient Reusable Nanocatalyst for Synthesis of Polyhydroquinolines in Water
1.1 Reagents: Ammonium acetate Catalysts: Copper (complex with Fe ₃ O ₄ -supported Schiff base), 2379757-15-8 (Fe ₃ O ₄ -supported, copper complex) Solvents: Water; 15 min, reflux		By: Ashraf, Muhammad Aqeel; et al Catalysis Letters (2020), 150(3), 683-701.
Experimental Protocols		
31-614-CAS-32531111	Steps: 1 Yield: 99%	Silica sulfuric acid coated on SnFe ₂ O ₄ MNPs: synthesis, characterization and catalytic applications in the synthesis of polyhydroquinolines
1.1 Reagents: Ammonium acetate Catalysts: Silica, Iron tin oxide (Fe ₂ SnO ₄) Solvents: Ethanol; 20 min, reflux		By: Esmaili, Soheila; et al RSC Advances (2022), 12(23), 14397-14410.
Experimental Protocols		

31-614-CAS-31172001 Steps: 1 Yield: 99% 1.1 Reagents: Ammonium acetate Catalysts: Hercynite (Al₂FeO₄) (sulfuric acid-modified); 20 min, 100 °C Experimental Protocols	Synthesis and characterization of novel hercynite@sulfuric acid and its catalytic applications in the synthesis of polyhydroquinolines and 2,3-dihydroquinazolin-4(1H)-ones By: Mohammadi, Masoud; et al RSC Advances (2022), 12(5), 2770-2787.
31-614-CAS-41856728 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: Iron oxide (Fe₃O₄), Silica, 3051706-75-0 Solvents: Ethanol; 20 min, reflux Experimental Protocols	The synthesis of new polyhydroquinoline derivatives using nano-[Fe₃O₄@SiO₂/N-propyl-1-(thiophene-2-yl) ethanimine] [CoCl₂] as a novel supported magnetite catalyst By: Gorji, Atefeh; et al Journal of Molecular Structure (2024), 1314, 138797.
31-614-CAS-39597935 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: 3025225-64-0 (silica-bound trimethoxysilane-aspergillomarasmine A-linked) Solvents: Ethanol; 15 min, reflux Experimental Protocols	Synthesis and characterization of a novel heterogenized six-coordinate aspergillomarasmine A-nickel(II) complex and its applications as an efficient nanomagnetic catalyst for Hantzsch reaction By: bayat, Vahid; et al Journal of Molecular Structure (2024), 1295(Part_1), 136635.
31-614-CAS-40737289 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: Tris[μ-[2-amino-1,4-benzenedicarboxylato(2-)-κO¹:κO¹]]diaquahydroxy-μ₃-oxotrichromium (post-treated with E DTA, Zn(OAc)₂) Solvents: Ethanol; 15 min, reflux Experimental Protocols	Post-Synthetic Modification of Amine-functionalized MIL-101 (Cr) Metal-Organic Frameworks with EDTA-Zn(II) Complex as an Effective Heterogeneous Catalyst for Hantzsch synthesis of Polyhydroquinolines By: Nikseresht, Ahmad; et al ACS Omega (2024), 9(26), 28114-28128.
31-614-CAS-35529727 Steps: 1 Yield: 98% 1.1 Catalysts: Silica, 1-Propanaminium, <i>N</i>-[2-(dimethylamino)ethyl]-<i>N,N</i>-dimethyl-3-(trimethoxysilyl)-, chloride (1:1); 10 min, 50 °C Experimental Protocols	Effectual and Rapid Synthesis of Hantzsch Derivatives in Solvent-Free Conditions Catalyzed by a Mesoporous Basic Silica-Based Nanomaterial By: Al Anazi, Abeer Abdullah; et al Silicon (2023), 15(8), 3453-3461.
31-614-CAS-33659247 Steps: 1 Yield: 98% 1.1 Catalysts: Carbon nitride (C₃N₄) Solvents: Ethanol; 20 min, reflux	1,4 Butanesultone Functionalized Graphitic Carbon Nitride: Efficient Catalysts for the One-Pot Synthesis of 1,4-Dihydropyridine and Polyhydroquinoline Derivative through Hantzsch Reaction By: Rahmati, Monavar; et al Polycyclic Aromatic Compounds (2022), 42(6), 3019-3035.
31-614-CAS-24071583 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: 2-(Hydroxymethyl)-2-[[3-(trimethoxysilyl)propyl]amino]-1,3-propanediol (Mesoporous SBA-15 supported; post-treated with ZrOCl₂) Solvents: Ethanol; 45 min, 80 °C Experimental Protocols	Mesoporous SBA-15@n-Pr-THAM-ZrO organic-inorganic hybrid: as a highly efficient reusable nanocatalyst for the synthesis of polyhydroquinolines and 2,3-dihydroquinazolin-4(1H)-ones By: Ghorbani-Choghamarani, Arash; et al Journal of Porous Materials (2021), 28(4), 1167-1186.

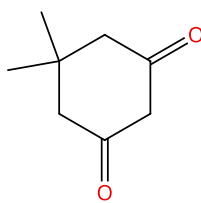
Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%



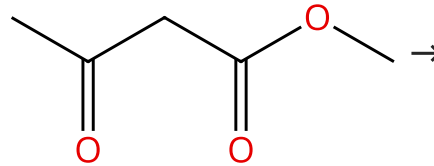
Suppliers (89)

+



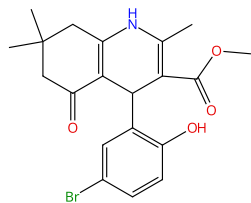
Suppliers (102)

+



Suppliers (101)

→



31-614-CAS-38151213

Steps: 1 Yield: 99%

1.1 Reagents: Ammonium acetate

Catalysts: 1,3,5-Benzenetricarboxylic acid (reaction products with cadmium and silver), Zinc oxide (ZnO), Silver(1+) (reaction products with 1,3,5-Benzenetricarboxylic acid), Cd²⁺ (reaction products with 1,3,5-Benzenetricarboxylic acid); 1 min, reflux

Experimental Protocols

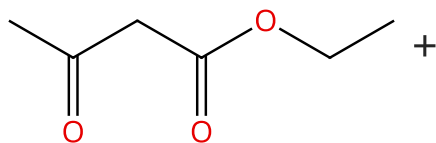
Synthesis and Characterization of Novel 2D Ultrathin Nanoribbons Cd-Ag-MOF@ZnO Composite as Organo-Nanocatalyst for Preparation of Polyhydroquinoline Heterocycles

By: Hoot, Negar; et al

Polycyclic Aromatic Compounds (2024), 44(8), 5218-5236.

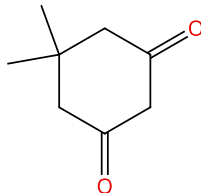
Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



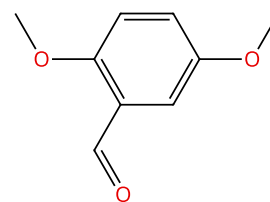
Suppliers (95)

+



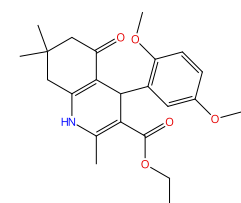
Suppliers (102)

+



Suppliers (86)

→



Suppliers (5)

31-614-CAS-39552451

Steps: 1 Yield: 99%

1.1 Reagents: Ammonium acetate

Catalysts: Papain (immobilized onto glutaraldehyde-functionalized nitrogen-doped graphene...), Graphene (papain immobilized onto glutaraldehyde-functionalized nitrogen-doped); 3 min

Experimental Protocols

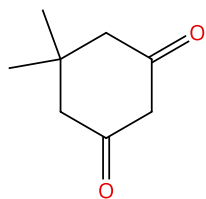
Immobilization of papain onto nitrogen graphene quantum dots and its use as an efficient catalyst for facile MW-assisted synthesis of polyhydroquinolines

By: Mirzaei, Farzaneh; et al

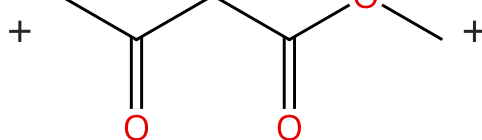
Journal of the Iranian Chemical Society (2024), 21(3), 639-648.

Scheme 7 (1 Reaction)

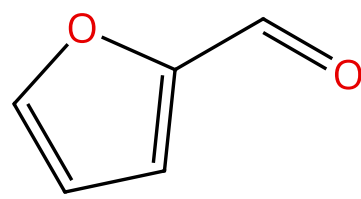
Steps: 1 Yield: 99%



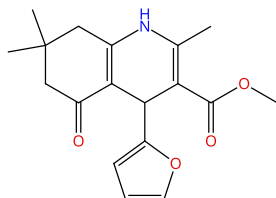
Suppliers (102)



Suppliers (101)



Suppliers (81)



31-614-CAS-38151207

Steps: 1 Yield: 99%

1.1 Reagents: Ammonium acetate

Catalysts: 1,3,5-Benzenetricarboxylic acid (reaction products with cadmium and silver), Zinc oxide (ZnO), Silver(1+) (reaction products with 1,3,5-Benzenetricarboxylic acid), Cd²⁺ (reaction products with 1,3,5-Benzenetricarboxylic acid); 6 min, reflux

Experimental Protocols

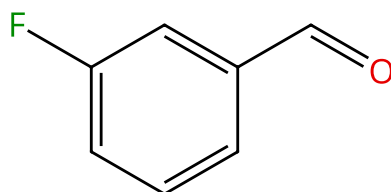
Synthesis and Characterization of Novel 2D Ultrathin Nanoribbons Cd-Ag-MOF@ZnO Composite as Organo-Nanocatalyst for Preparation of Polyhydroquinoline Heterocycles

By: Hoot, Negar; et al

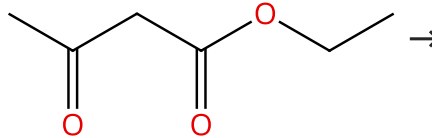
Polycyclic Aromatic Compounds (2024), 44(8), 5218-5236.

Scheme 8 (1 Reaction)

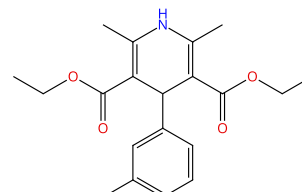
Steps: 1 Yield: 99%



Suppliers (82)



Suppliers (95)



Suppliers (4)

31-614-CAS-43166720

Steps: 1 Yield: 99%

1.1 Reagents: Ammonium bicarbonate

Catalysts: 3068689-09-5

Solvents: Ethanol; 24 h, 80 °C

Experimental Protocols

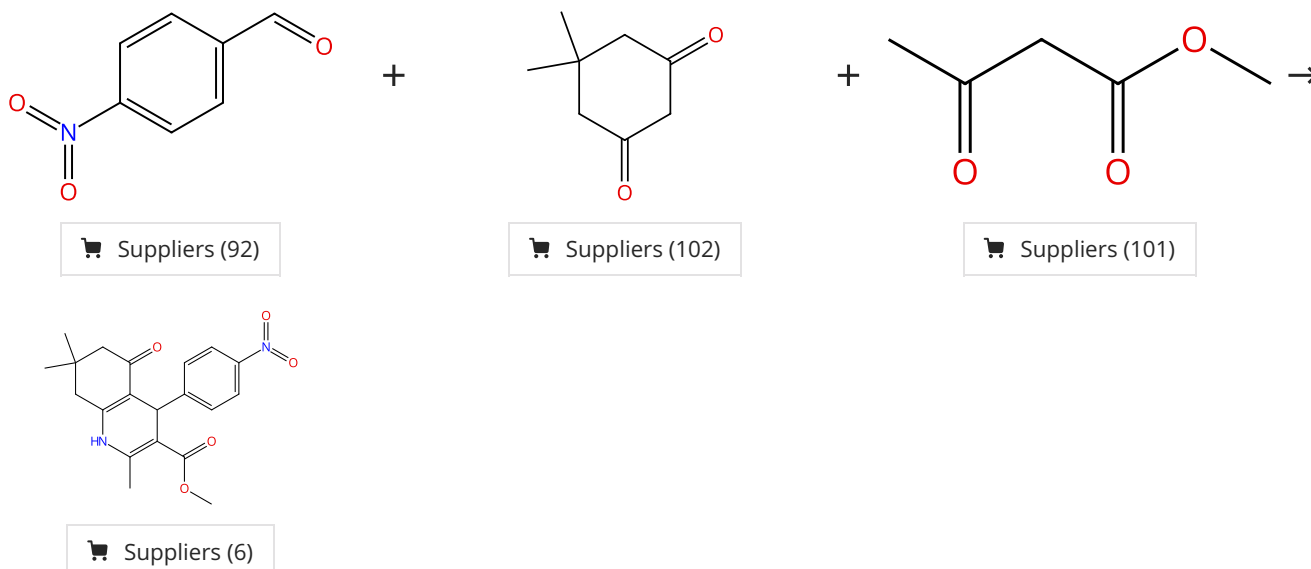
2D Cerium-Organic Frameworks as an Efficient Heterogeneous Catalyst for the Synthesis of 1,4-Dihydropyridines via Hantzsch Reaction

By: Yu, Yi-Han; et al

Chemistry - An Asian Journal (2025), 20(3), e202400977.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-38151206

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Ammonium acetate
Catalysts: 1,3,5-Benzenetricarboxylic acid (reaction products with cadmium and silver), Zinc oxide (ZnO), Silver(1+) (reaction products with 1,3,5-Benzenetricarboxylic acid), Cd²⁺ (reaction products with 1,3,5-Benzenetricarboxylic acid); 1 min, reflux

Experimental Protocols

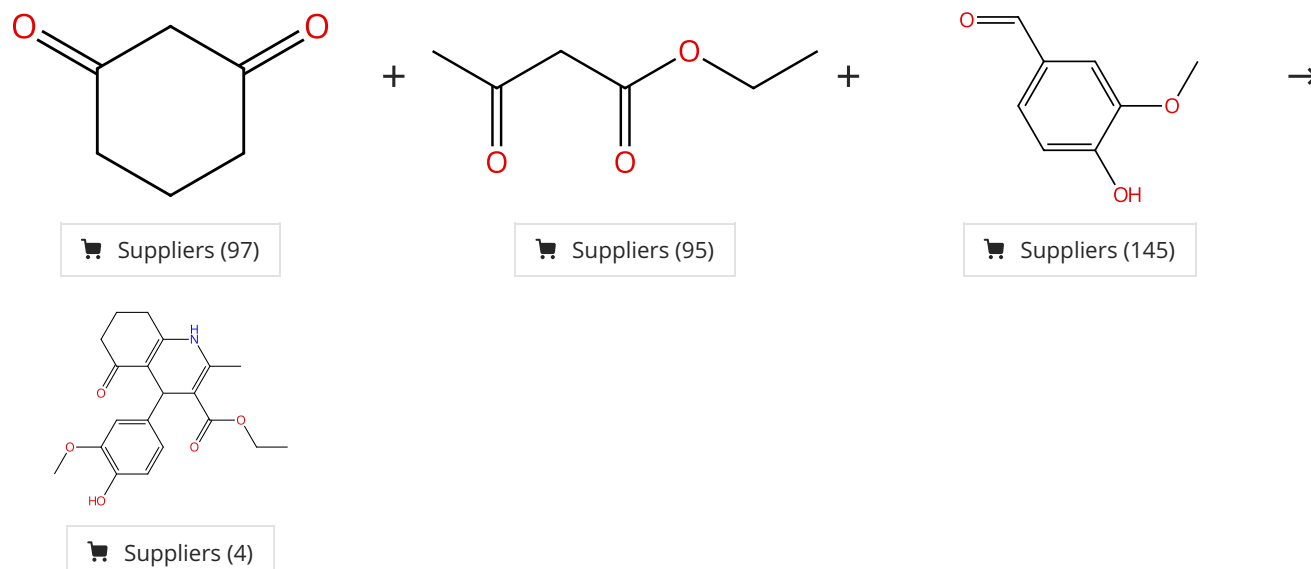
Synthesis and Characterization of Novel 2D Ultrathin Nanoribbons Cd-Ag-MOF@ZnO Composite as Organo-Nanocatalyst for Preparation of Polyhydroquinoline Heterocycles

By: Hoot, Negar; et al

Polycyclic Aromatic Compounds (2024), 44(8), 5218-5236.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-37804057

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Ammonium acetate
Catalysts: 2985010-79-3; 2 min, 120 °C

Experimental Protocols

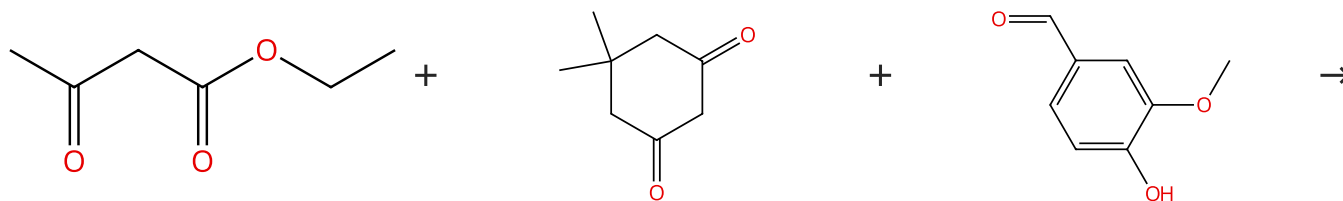
One-pot synthesis of 1,8-dioxo-decahydroacridine and hexahydroquinoline derivatives catalyzed by a new natural deep eutectic mixture

By: Rahro, Parissa Naddaf; et al

Research on Chemical Intermediates (2023), 49(11), 4879-4895.

Scheme 11 (1 Reaction)

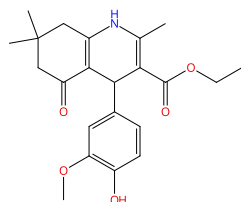
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (102)

Suppliers (145)



Suppliers (7)

31-614-CAS-37804032

Steps: 1 Yield: 99%

One-pot synthesis of 1,8-dioxo-decahydroacridine and hexahydroquinoline derivatives catalyzed by a new natural deep eutectic mixture

1.1 Reagents: Ammonium acetate
Catalysts: 2985010-79-3; 4 min, 120 °C

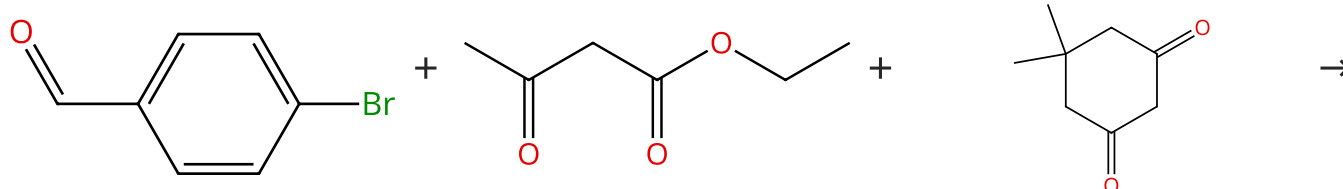
By: Rahro, Parissa Naddaf; et al

Experimental Protocols

Research on Chemical Intermediates (2023), 49(11), 4879-4895.

Scheme 12 (1 Reaction)

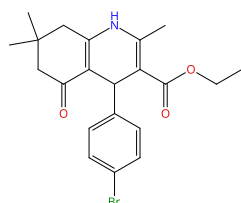
Steps: 1 Yield: 99%



Suppliers (91)

Suppliers (95)

Suppliers (102)



Suppliers (15)

31-614-CAS-45521447

Steps: 1 Yield: 99%

DABCO-modified magnetic core-shell as an efficient nanocatalyst for synthesizing polyhydroquinoline derivatives

1.1 Reagents: Ammonium acetate
Catalysts: Iron oxide (Fe₃O₄), 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-[3-(trimethoxysilyl)propyl]-, chloride (1:1) (RF coated ferric oxide-supported); 5 min, 60 °C

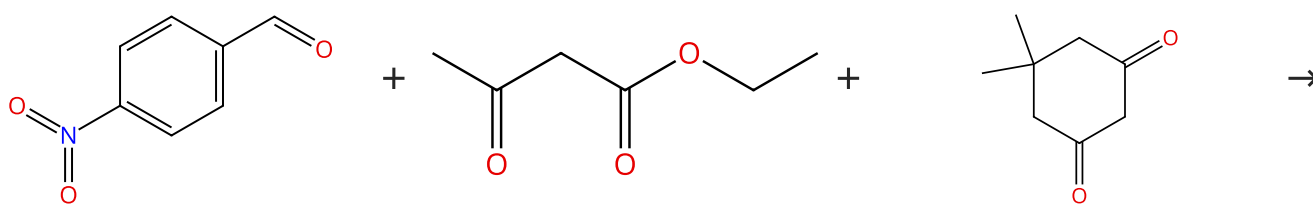
By: Esfandiari, Mozghan; et al

Frontiers in Chemistry (Lausanne, Switzerland) (2025), 13, 1557628.

Experimental Protocols

Scheme 13 (1 Reaction)

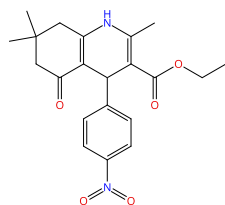
Steps: 1 Yield: 99%



Suppliers (92)

Suppliers (95)

Suppliers (102)



Suppliers (9)

31-614-CAS-39552440

Steps: 1 Yield: 99%

Immobilization of papain onto nitrogen graphene quantum dots and its use as an efficient catalyst for facile MW-assisted synthesis of polyhydroquinolines

By: Mirzaei, Farzaneh; et al

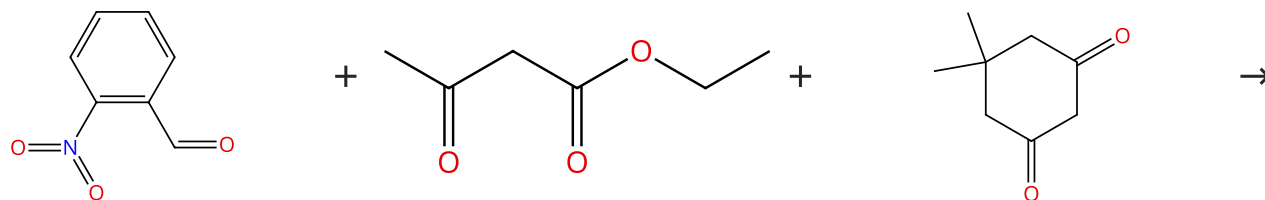
Journal of the Iranian Chemical Society (2024), 21(3), 639-648.

1.1 **Reagents:** Ammonium acetate
Catalysts: Papain (immobilized onto glutaraldehyde-functionalized nitrogen-doped graphene...), Graphene (papain immobilized onto glutaraldehyde-functionalized nitrogen-doped); 3 min

Experimental Protocols

Scheme 14 (1 Reaction)

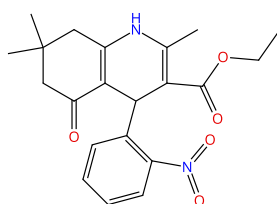
Steps: 1 Yield: 99%



Suppliers (99)

Suppliers (95)

Suppliers (102)



Suppliers (7)

31-614-CAS-37804039

Steps: 1 Yield: 99%

One-pot synthesis of 1,8-dioxo-decahydroacridine and hexahydroquinoline derivatives catalyzed by a new natural deep eutectic mixture

By: Rahro, Parissa Naddaf; et al

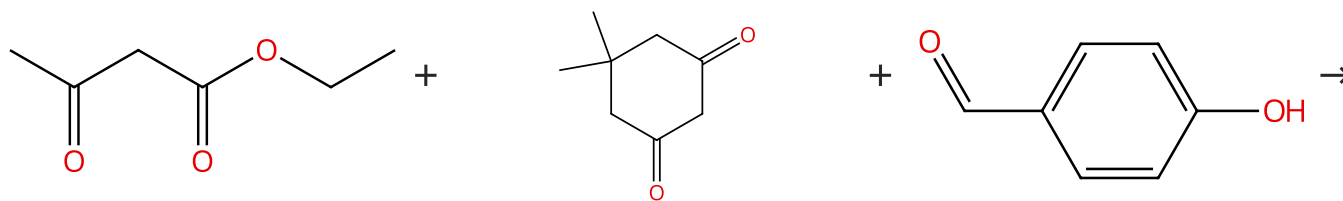
Research on Chemical Intermediates (2023), 49(11), 4879-4895.

1.1 **Reagents:** Ammonium acetate
Catalysts: 2985010-79-3; 14 min, 120 °C

Experimental Protocols

Scheme 15 (1 Reaction)

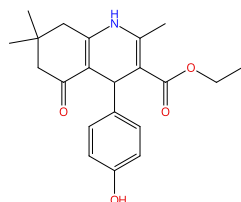
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (102)

Suppliers (121)



Suppliers (11)

31-614-CAS-37804004

Steps: 1 Yield: 99%

One-pot synthesis of 1,8-dioxo-decahydroacridine and hexahydroquinoline derivatives catalyzed by a new natural deep eutectic mixture

1.1 **Reagents:** Ammonium acetate
Catalysts: 2985010-79-3; 4 min, 120 °C

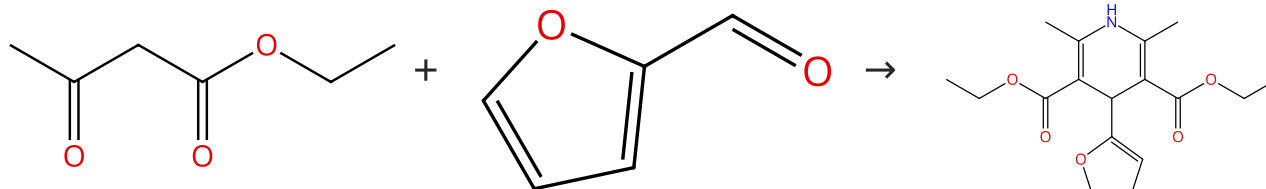
By: Rahro, Parissa Naddaf; et al

Experimental Protocols

Research on Chemical Intermediates (2023), 49(11), 4879-4895.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (81)

Suppliers (19)

31-614-CAS-43166725

Steps: 1 Yield: 99%

2D Cerium-Organic Frameworks as an Efficient Heterogeneous Catalyst for the Synthesis of 1,4-Dihydropyridines via Hantzsch Reaction

1.1 **Reagents:** Ammonium bicarbonate
Catalysts: 3068689-09-5
Solvents: Ethanol; 24 h, 80 °C

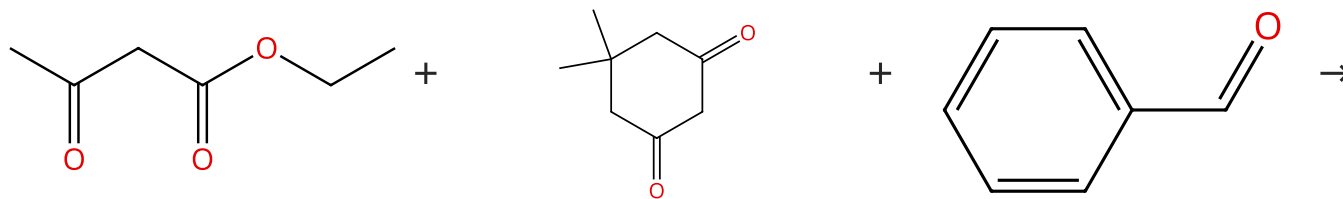
By: Yu, Yi-Han; et al

Experimental Protocols

Chemistry - An Asian Journal (2025), 20(3), e202400977.

Scheme 17 (7 Reactions)

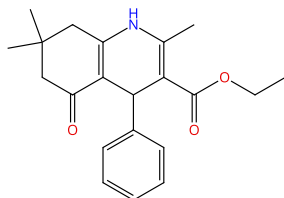
Steps: 1 Yield: 98-99%



Suppliers (95)

Suppliers (102)

Suppliers (69)



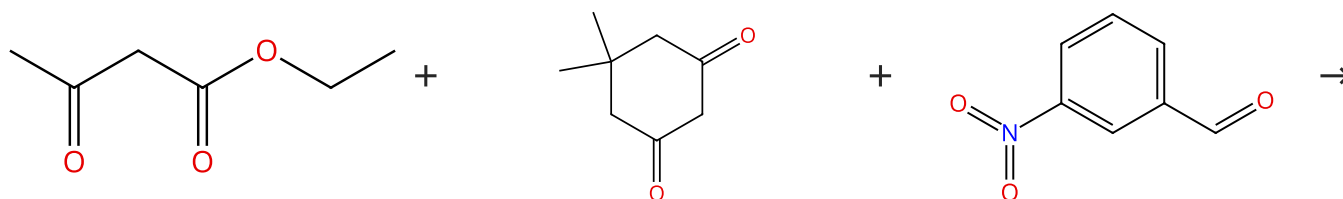
Suppliers (14)

31-614-CAS-39552442 Steps: 1 Yield: 99% 1.1 Reagents: Ammonium acetate Catalysts: Papain (immobilized onto glutaraldehyde-functionalized nitrogen-doped graphene...), Graphene (papain immobilized onto glutaraldehyde-functionalized nitrogen-doped); 3 min Experimental Protocols	Immobilization of papain onto nitrogen graphene quantum dots and its use as an efficient catalyst for facile MW-assisted synthesis of polyhydroquinolines By: Mirzaei, Farzaneh; et al Journal of the Iranian Chemical Society (2024), 21(3), 639-648.
31-614-CAS-32531104 Steps: 1 Yield: 99% 1.1 Reagents: Ammonium acetate Catalysts: Silica, Iron tin oxide (Fe_2SnO_4) Solvents: Ethanol; 30 min, reflux Experimental Protocols	Silica sulfuric acid coated on SnFe_2O_4 MNPs: synthesis, characterization and catalytic applications in the synthesis of polyhydroquinolines By: Esmaili, Soheila; et al RSC Advances (2022), 12(23), 14397-14410.
31-333-CAS-21579516 Steps: 1 Yield: 99% 1.1 Reagents: Ammonium acetate Catalysts: Copper (complex with Fe_3O_4-supported Schiff base), 2379757-15-8 (Fe_3O_4-supported, copper complex) Solvents: Water; 20 min, reflux Experimental Protocols	New Copper Complex on Fe_3O_4 Nanoparticles as a Highly Efficient Reusable Nanocatalyst for Synthesis of Polyhydroquinolines in Water By: Ashraf, Muhammad Aqeel; et al Catalysis Letters (2020), 150(3), 683-701.
31-614-CAS-35529723 Steps: 1 Yield: 98% 1.1 Catalysts: Silica, 1-Propanaminium, <i>N</i>-[2-(dimethylamino)ethyl]-<i>N,N</i>-dimethyl-3-(trimethoxysilyl)-, chloride (1:1); 9 min, 50 °C Experimental Protocols	Effectual and Rapid Synthesis of Hantzsch Derivatives in Solvent-Free Conditions Catalyzed by a Mesoporous Basic Silica-Based Nanomaterial By: Al Anazi, Abeer Abdullah; et al Silicon (2023), 15(8), 3453-3461.
31-614-CAS-32398417 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: Iron oxide (Fe_3O_4), Boron trifluoride, Dextrin; 20 min, 80 °C Experimental Protocols	One-pot synthesis of polyhydroquinoline derivatives using nano-Fe_3O_4@dextrin/BF_3 as a magnetic biodegradable catalyst By: Karimi Askarani, Hajar; et al Journal of the Iranian Chemical Society (2022), 19(7), 3189-3203.

31-614-CAS-31172004 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: Hercynite (Al₂FeO₄) (sulfuric acid-modified); 30 min, 100 °C Experimental Protocols	Synthesis and characterization of novel hercynite@sulfuric acid and its catalytic applications in the synthesis of polyhydroquinolines and 2,3-dihydroquinazolin-4(1H)-ones By: Mohammadi, Masoud; et al RSC Advances (2022), 12(5), 2770-2787.
31-333-CAS-22732035 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: Poly(vinylpyrrolidone) (sulfuric acid-modified); 8 min, 110 °C Experimental Protocols	Application of [PVP-SO₃H] HSO₄ as Powerful Polymeric-Based Solid Acid Catalyst for Hantzsch Synthesis of Polyhydroquinolin-5(1H)-One By: Pakpour-Roudsari, Fatemeh; et al Polycyclic Aromatic Compounds (2020), 40(5), 1321-1331.

Scheme 18 (1 Reaction)

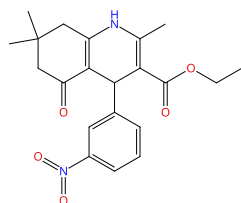
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (102)

Suppliers (90)

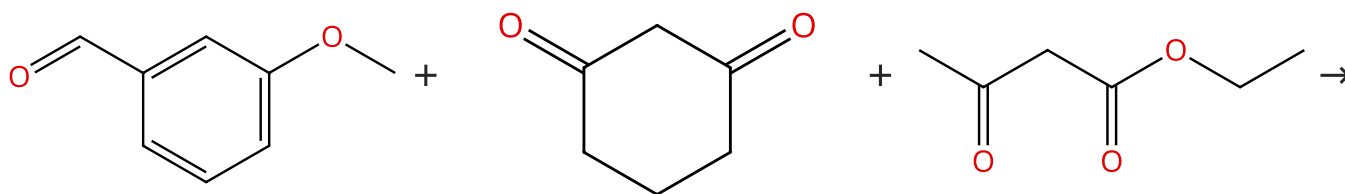


Suppliers (6)

31-614-CAS-39552439 Steps: 1 Yield: 99% 1.1 Reagents: Ammonium acetate Catalysts: Papain (immobilized onto glutaraldehyde-functionalized nitrogen-doped graphene...), Graphene (papain immobilized onto glutaraldehyde-functionalized nitrogen-doped); 3 min Experimental Protocols	Immobilization of papain onto nitrogen graphene quantum dots and its use as an efficient catalyst for facile MW-assisted synthesis of polyhydroquinolines By: Mirzaei, Farzaneh; et al Journal of the Iranian Chemical Society (2024), 21(3), 639-648.
---	---

Scheme 19 (1 Reaction)

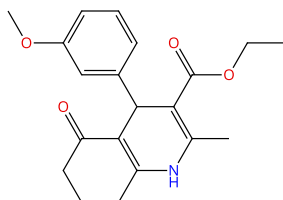
Steps: 1 Yield: 99%



Suppliers (89)

Suppliers (97)

Suppliers (95)



Suppliers (7)

31-614-CAS-37804047

Steps: 1 Yield: 99%

One-pot synthesis of 1,8-dioxo-decahydroacridine and hexahydroquinoline derivatives catalyzed by a new natural deep eutectic mixture

1.1 Reagents: Ammonium acetate
Catalysts: 2985010-79-3; 5 min, 120 °C

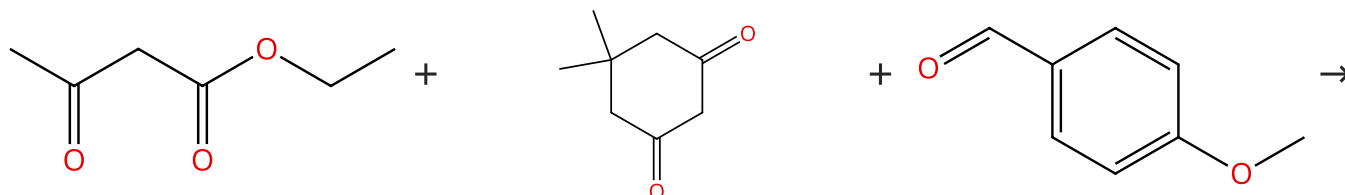
Experimental Protocols

By: Rahro, Parissa Naddaf; et al

Research on Chemical Intermediates (2023), 49(11), 4879-4895.

Scheme 20 (2 Reactions)

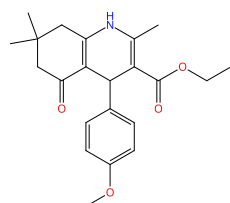
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (102)

Suppliers (128)



Suppliers (22)

31-614-CAS-45521449

Steps: 1 Yield: 99%

DABCO-modified magnetic core-shell as an efficient nanocatalyst for synthesizing polyhydroquinoline derivatives

1.1 Reagents: Ammonium acetate
Catalysts: Iron oxide (Fe₃O₄), 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-[3-(trimethoxysilyl)propyl]-, chloride (1:1) (RF coated ferric oxide-supported); 10 min, 60 °C

Experimental Protocols

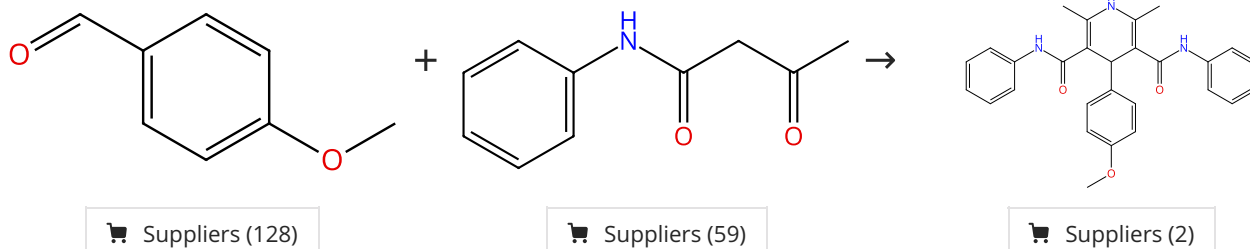
By: Esfandiari, Mozghan; et al

Frontiers in Chemistry (Lausanne, Switzerland) (2025), 13, 1557628.

31-614-CAS-31085197	Steps: 1 Yield: 99%	Magnetic core-shell dendritic mesoporous silica nanospheres anchored with diamine as an efficient and recyclable base catalyst
1.1 Reagents: Ammonium acetate Catalysts: Iron oxide (Fe ₃ O ₄), <i>N</i> -(2-Aminoethyl)-3-aminopropyltrimethoxysilane (surface-bound), 1-Hexadecanaminium, <i>N,N,N</i> -triethyl-, chloride (1:1) Solvents: Water; 12 min, 35 °C		By: Surabhi; et al New Journal of Chemistry (2020), 44(48), 21152-21166.
Experimental Protocols		

Scheme 21 (1 Reaction)

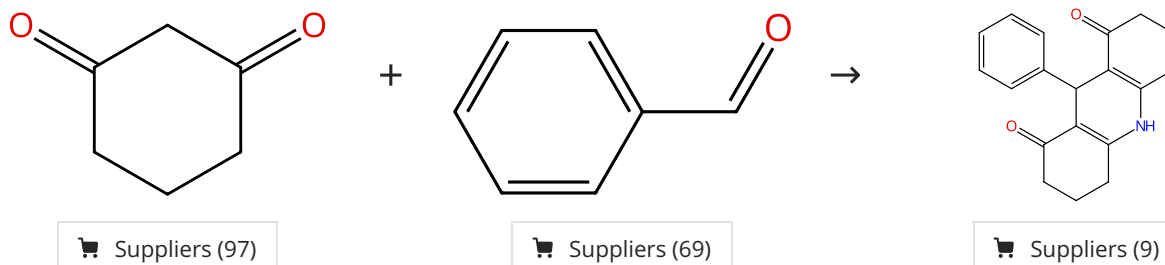
Steps: 1 Yield: 98%



31-333-CAS-22617886	Steps: 1 Yield: 98%	Magnetic Fe-C-O-Mo alloy nano-rods prepared from chemical decomposition of a screw (a top-down approach): An efficient and cheap catalyst for the preparation of dihydropyridine and dihydropyrimidone derivatives
1.1 Reagents: Ammonium acetate Catalysts: Carbon iron molybdenum oxide Solvents: Ethanol; 40 min, reflux		By: Wu, Pei; et al Applied Catalysis, A: General (2020), 590, 117301.
Experimental Protocols		

Scheme 22 (1 Reaction)

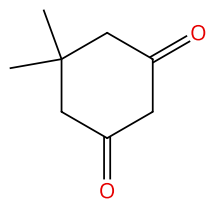
Steps: 1 Yield: 98%



31-614-CAS-37803883	Steps: 1 Yield: 98%	One-pot synthesis of 1,8-dioxo-decahydroacridine and hexahydroquinoline derivatives catalyzed by a new natural deep eutectic mixture
1.1 Reagents: Ammonium acetate Catalysts: 2985010-79-3; 3 min, 120 °C		By: Rahro, Parissa Naddaf; et al Research on Chemical Intermediates (2023), 49(11), 4879-4895.
Experimental Protocols		

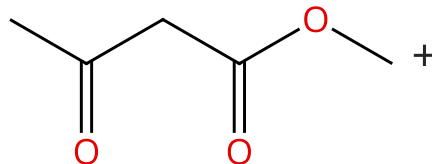
Scheme 23 (1 Reaction)

Steps: 1 Yield: 98%



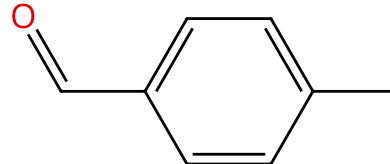
Suppliers (102)

+



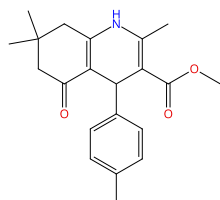
Suppliers (101)

+



Suppliers (104)

→



Suppliers (5)

31-614-CAS-33997655

Steps: 1 Yield: 98%

Introduction of a tropine-based dication ionic liquid catalyst for the synthesis of polyhydroquinoline and 1,8-dioxodec ahydroacridine derivatives

1.1 Reagents: Ammonium acetate
Catalysts: 2832006-99-0; 12 min, 120 °C

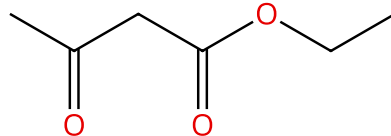
By: Pourdasht, Sepideh; et al

Experimental Protocols

Research on Chemical Intermediates (2022), 48(10), 4403-4418.

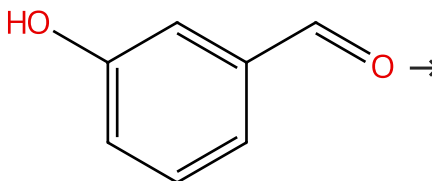
Scheme 24 (1 Reaction)

Steps: 1 Yield: 98%



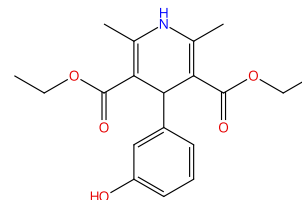
Suppliers (95)

+



Suppliers (96)

→



Suppliers (10)

31-614-CAS-37220325

Steps: 1 Yield: 98%

A CMC-g-poly(AA-co-AMPS)/Fe₃O₄ hydrogel nanocomposite as a novel biopolymer-based catalyst in the synthesis of 1,4-dihydropyridines

1.1 Reagents: Ammonium acetate
Catalysts: Iron oxide (Fe₃O₄), Cellulose, carboxymethyl ether, sodium salt, polymer with 2-methyl-2-[(1-oxo-2-propen-1-yl) amino]-1-propanesulfonic acid and 2-propenoic acid, graft; 10 h, 40 °C

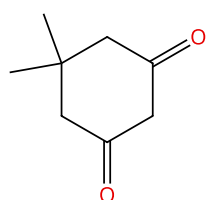
By: Hemmati, Elmira; et al

RSC Advances (2023), 13(24), 16567-16583.

Experimental Protocols

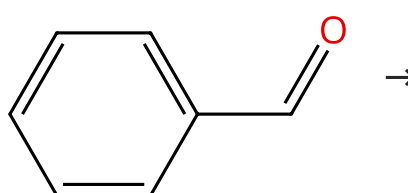
Scheme 25 (2 Reactions)

Steps: 1 Yield: 98%



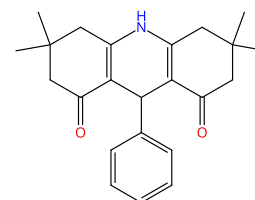
Suppliers (102)

+



Suppliers (69)

→

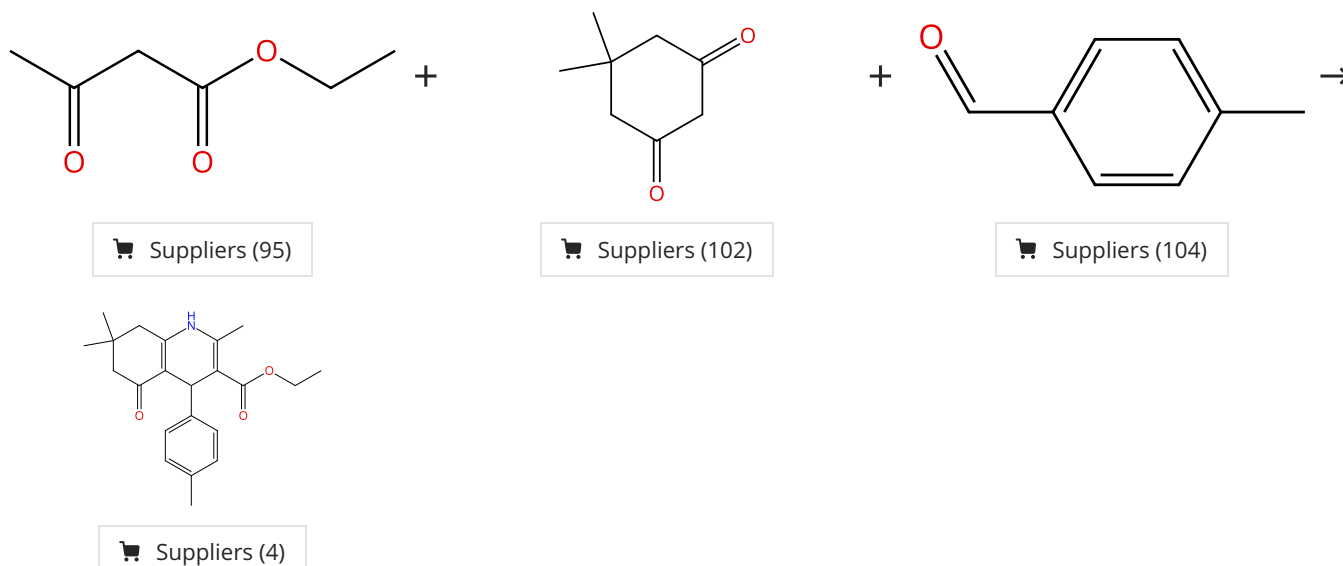


Suppliers (9)

31-614-CAS-35529740 Steps: 1 Yield: 98% 1.1 Catalysts: Silica, 1-Propanaminium, <i>N</i>-[2-(dimethylamino)ethyl]-<i>N,N</i>-dimethyl-3-(trimethoxysilyl)-, chloride (1:1); 8 min, 50 °C Experimental Protocols	Effectual and Rapid Synthesis of Hantzsch Derivatives in Solvent-Free Conditions Catalyzed by a Mesoporous Basic Silica-Based Nanomaterial By: Al Anazi, Abeer Abdullah; et al Silicon (2023), 15(8), 3453-3461.
31-329-CAS-23767686 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: β-Cyclodextrin (reaction products with imidazole and chlorosilane), 1,1'-(1,4-Butanediyl)bis(imidazole) (reaction products with cyclodextrin and chlorosilane), [3-Chloro-2-(chloromethyl)propyl]trimethoxysilane (reaction products with cyclodextrin and imidazole); 15 min, 90 °C Experimental Protocols	Synthesis, Characterization and Application of β-Cyclodextrin/Imidazolium Based Dicationic Ionic Liquid Supported on Silica Gel as a Novel Catalyst in Hantzsch Condensation Reaction By: Moheiseni, Fatemeh; et al Polycyclic Aromatic Compounds (2021), 41(5), 1094-1106.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-31085214 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium acetate Catalysts: Iron oxide (Fe₃O₄), <i>N</i>-(2-Aminoethyl)-3-aminopropyltrimethoxysilane (surface-bound), 1-Hexadecanaminium, <i>N,N,N</i>-triethyl-, chloride (1:1) Solvents: Water; 15 min, 35 °C Experimental Protocols	Magnetic core-shell dendritic mesoporous silica nanospheres anchored with diamine as an efficient and recyclable base catalyst By: Surabhi; et al New Journal of Chemistry (2020), 44(48), 21152-21166.
--	---